UNIT I
BUILDING MATERIALS

- **STONES**
- **BRICKS**
- **CEMENT**
- **CONCRETE**
- ***STEEL***











Early Use of Metals

- Iron has been used since 4000 BC, at least. The Iron Age continues even today with the widespread use of iron and steel.
- High quality steel was produced in southern India perhaps as early as 300 BC.



The Iron Pillar of Delhi, 402 BC

"A testament to the skill of ancient Indian blacksmiths" because of its high resistance to corrosion

Non-ferrous and ferrous metal

Non- Ferrous Metals

Copper, Aluminum, Zinc, Lead, Tin, Titanium, Gold, Silver, etc.

Ferrous Metals

- A ferrous material is the one in which iron is a main constituent.
- processes to mix different **percentage of carbon** and to get the following three useful ferrous

Materials:

1. Cast Iron
2. Wrought Iron
3. Steel

- All ferrous materials contain about **0.5 to 3% silica**, less than **2% manganese**, **0.15% sulphur** and **0.6% phosphorous.**

Types of Iron

Description		Cast Iron	Wrought Iron	Steel
Carbon Content		1.7 – 4.5 %	< 0.15 %	0.25 - 1.5%.
Strength (N/mm^2)	**Compression**	700	200	
	Tension	150	375	Fe – 250; Fe – 415 & 500
Nature		Brittle, Not shock absorbs	Ductile & Brittle Absorb shock	Ductile & Malleable Absorb shock
Magnetize Property		Not magnetized	Temporarily magnetize	Permanently magnetize
Rust property		Not rust	Rust	Rust
Melting point		1200°C		
Uses		•Rain water •Sanitary pipes •Sanitary fittings •Manhole covers. •Railings •Spiral stair cases •Fire gratings • Cover for pumps	•Nails, nuts and bolts •Wires and chains • Roofing sheets •Grills and fences •Window guards etc.	• Mild steel – 0.25 % Carbon • High carbon steel – 0.7 – 1.5 % Carbon •High tensile steel- 0.8 % carbon & 0.6 % manganese.

Steel

- Steel is probably the most versatile commonly used structural material.
- Steel is used to a large extent in modern multi-storied buildings.
- Steel is used as reinforcing bars/wires for concrete since concrete is weak in tension.
- Structural steel is available in various forms and shapes and it is being used for various structural components.

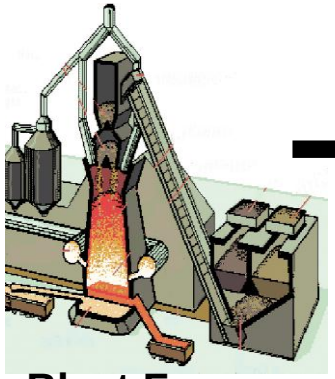




How is steel manufactured?



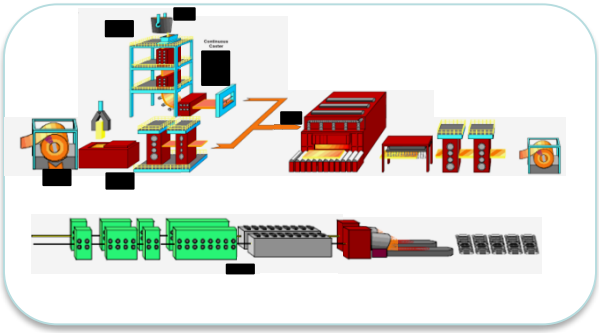
Iron Ore
(Fe_2O_3)



Blast Furnace



Molten material



Rolling/ other manufacturing processes

Steel is an alloy of iron and carbon

- Steel is obtained by decreasing the carbon content by controlled oxidation. The excess oxygen is removed by incorporating manganese and silicon.
- Classification of steels:
  - Mild steel or low carbon steel: carbon content of upto 0.25%.
  - Medium carbon steel or carbon steel: C content of 0.3-0.6%.
  - High carbon steels: C content of more than 0.6%.
  - When elements other than Mn and Si are present, it is called alloy steel.
  - When certain elements, such as Cr and Ni are added, it is called stainless steel.

Mild steel

It contains a maximum of 0.25% carbon, 0.055% of sulphur and 0.55% of phosphorus.

Properties of Mild Steel:

- It is malleable and ductile. It is more elastic
- It can be magnetized permanently.
- Its specific gravity is 7.8.
- Its Young's modulus is 2.1×10^5 N/mm^2 or GPa
- It can be welded easily.
- It is equally strong in tension and in compression.

Uses of Mild Steel

- Round bars are extensively used as reinforcement in R.C.C. works.
- Rolled sections like I, T, L, C, plates etc. are used to build steel columns, beams, trusses etc.
- Tubular sections are used as poles and members of trusses.
- Plain and corrugated mild steel are used as roofing materials.
- Mild steel sections are used in making parts of many machineries.

Types of reinforcement

- Uncoated steel
  - Mild steel ribbed bars
  - High Yield Strength Deformed (HYSD) bars
  - Cold-twisted deformed (CTD) bars (TORsteel)
  - Thermomechanically treated (TMT) or Quenched and Self-Tempered (QST) steel
  - Stainless steel
  - Prestressing steel
- Coated steel
  - Epoxy coated steel
  - Galvanized steel
- Fiber-reinforced Polymer (FRP) bars / laminates

Nominal area and mass of reinforcement used in India – IS 1786

Sl No.	Nominal Size mm	Cross-Sectional Area mm^2	Mass per Metre kg
(1)	(2)	(3)	(4)
i)	4	12.6	0.099
ii)	5	19.6	0.154
iii)	6	28.3	0.222
iv)	8	50.3	0.395
v)	10	78.6	0.617
vi)	12	113.1	0.888
vii)	16	201.2	1.58
viii)	20	314.3	2.47
ix)	25	491.1	3.85
x)	28	615.8	4.83
xi)	32	804.6	6.31
xii)	36	1 018.3	7.99
xiii)	40	1 257.2	9.86

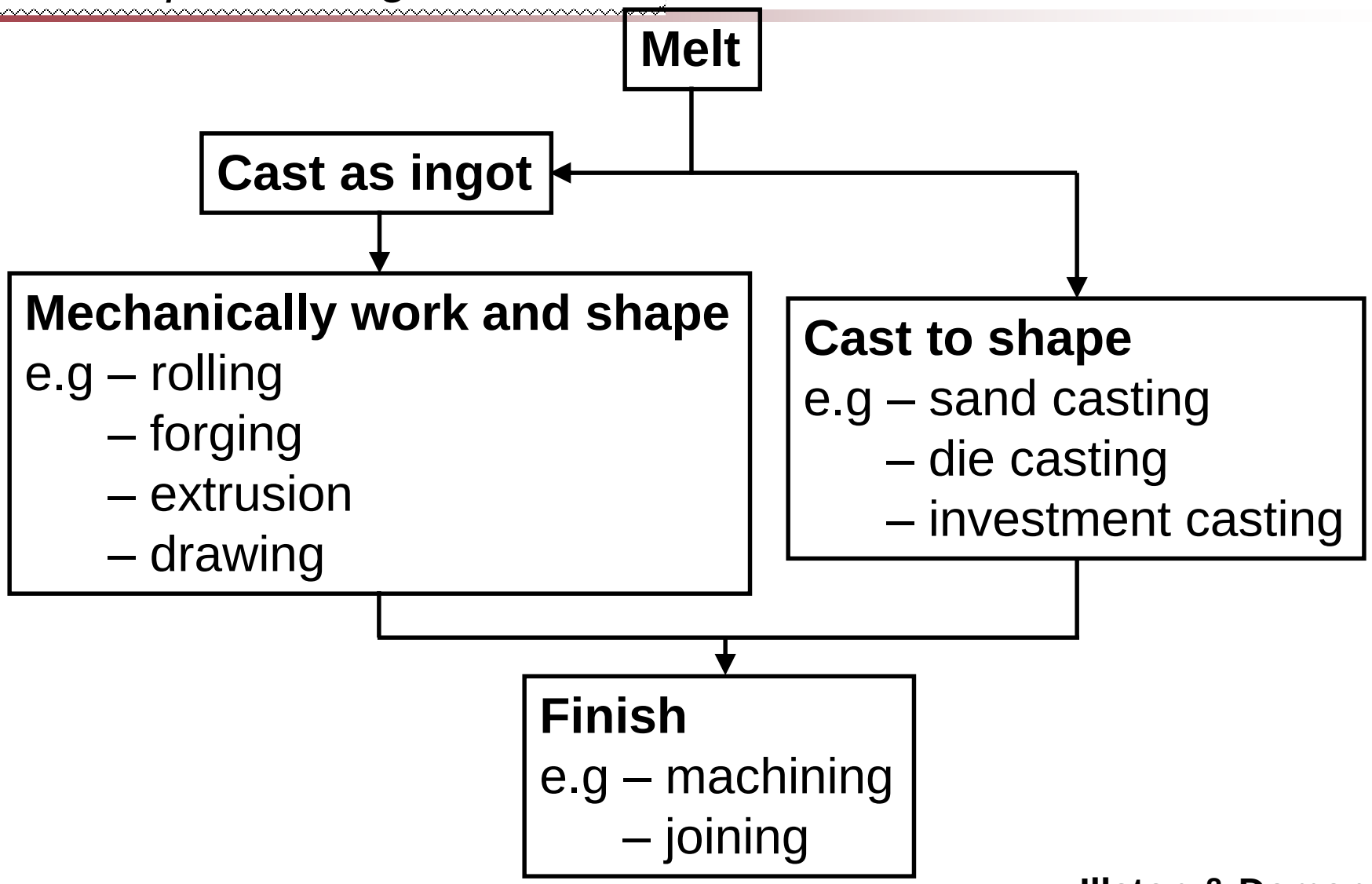
Mechanical properties of some typical reinforcement used in India – IS 1786

- Grades Fe415, Fe500 and Fe550
  - –the number indicates the yield stress
  - –Grade Fe250 mild steel is also available but used only as secondary reinforcement.

		Limiting value		
S. No	Property	Fe 415 Grade	Fe 500 Grade	Fe 550 Grade
1.	0.2% proof stress/ yield stress, min, mpa	415	500	550
2.	Elongation, percent, min, on guage length 5.65$\sqrt{A}$, where is the cross sectional area of the test piece	14.5	12.0	8.0
3.	Tensile strength. min	10% more than actual 0.2% proof stress but not less than 485 Mpa	8% more than actual 0.2% proof stress but not less than 545 Mpa	6% more than actual 0.2% proof stress but not less than 585 Mpa

Forming of Metals

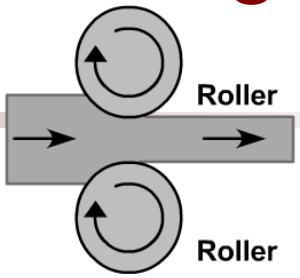
Common processing methods



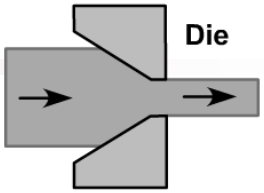
Illston & Domone

Forming of Metals - Hot working

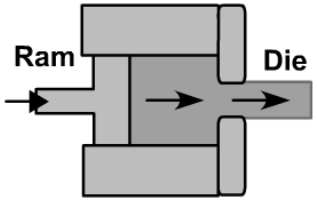
Hot working processes depend on plasticity, which is higher at high temperatures, i.e., above the recrystallisation temperature. This allows common metals to be deformed in compression, without breaking.



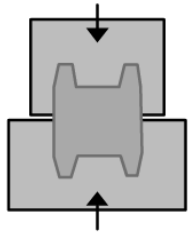
Rolling

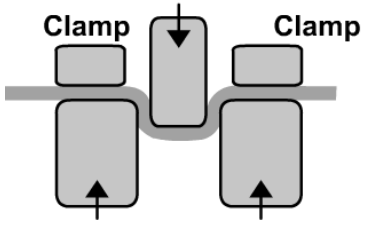


Drawing

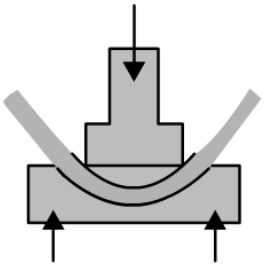


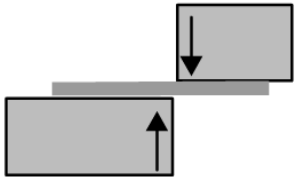
Extrusion

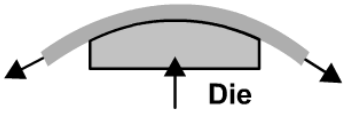

Forging



Deep drawing


Bending


Shearing



Stretch forming

Forming of Metals

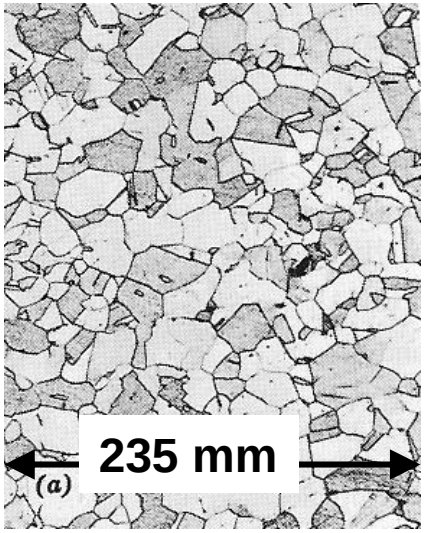
- For steel structural members, the usual method is by *hot rolling*, with cylindrical or shaped rollers at temperatures of 1000 °C or more. The members are later allowed to cool naturally, ending up with an annealed microstructure. The exposure of the hot metal to air causes a heavy film of oxide on the surface.

- One disadvantage of hot forming arises from the contraction of the dimensions on cooling.

Forming of Metals - Cold working

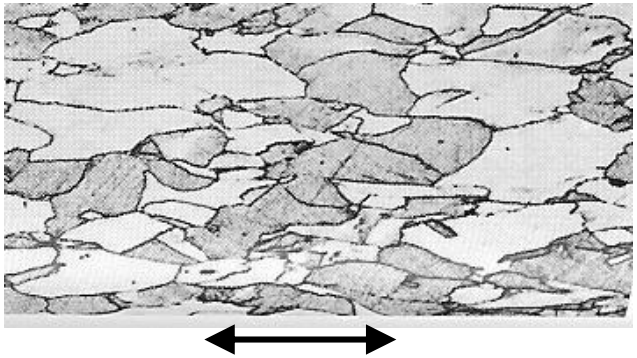
- Many metals and alloys can be cold worked, i.e., shaped at temperatures below that of recrystallisation.
- This creates an immense amount dislocations, and consequently, work hardening. As a result the yield strength increases.
- Cold working processes can produce anisotropy in polycrystalline metals due to the deformation of the grains.

before rolling



Isotropic – equiaxed grains

after rolling



rolling direction

Anisotropic – elongated and oriented grains

Cold-Twisted Deformed (CTD) bars - TOR steel

- CTD bars are produced by twisting in cold condition the steel bars, causing work hardening.
  - Higher strength and lower elongation
  - Ribs improve the bonding with the concrete

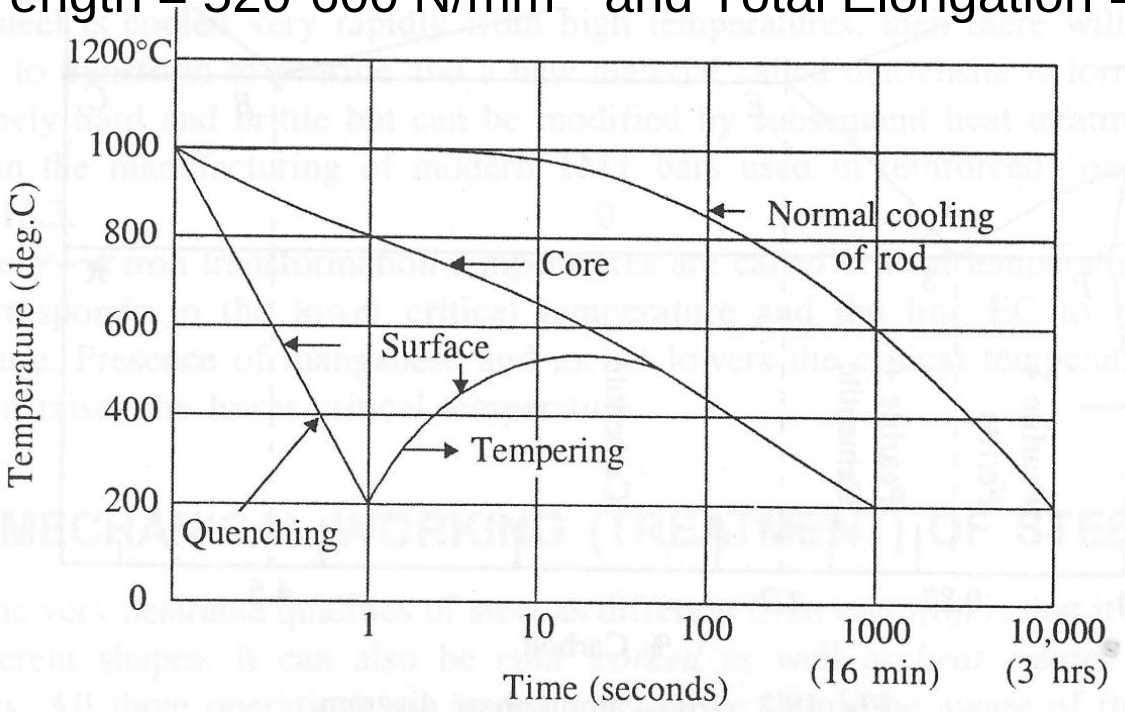


- Chemical Composition
  - Carbon: 0.20-0.35%
  - Manganese: 0.40-0.80%
  - Phosphorus & Sulphur combined: 0.120% max

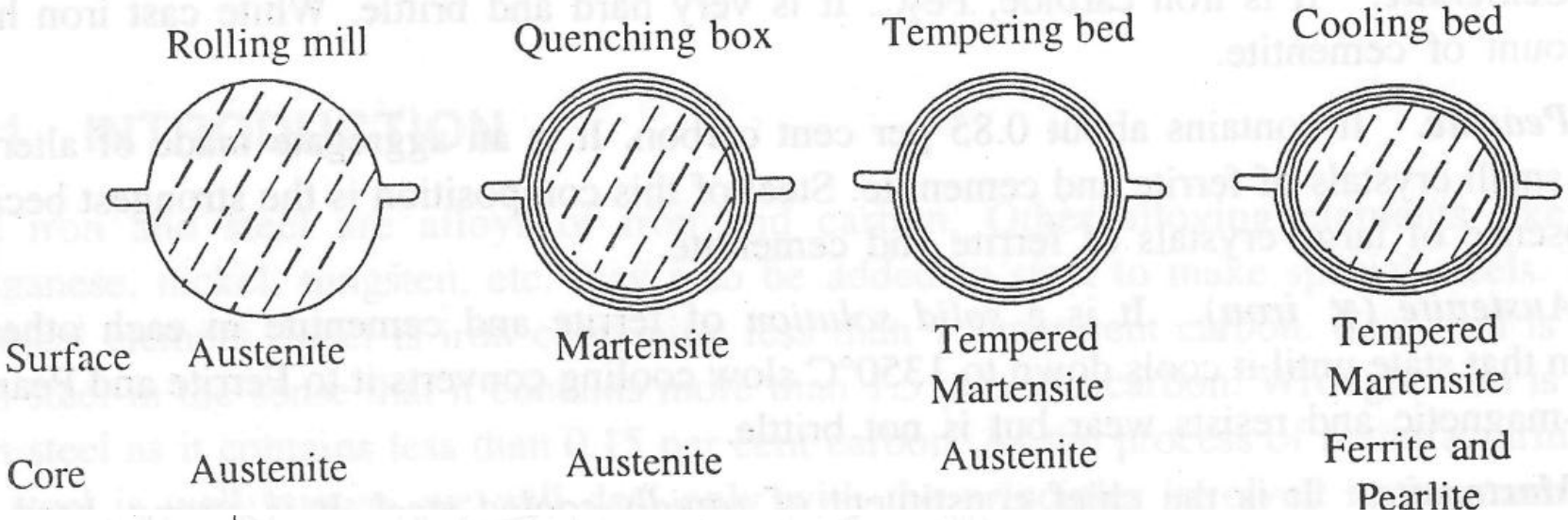
- CTD bars are no longer in use due to poor corrosion resistance and stress corrosion cracking issues

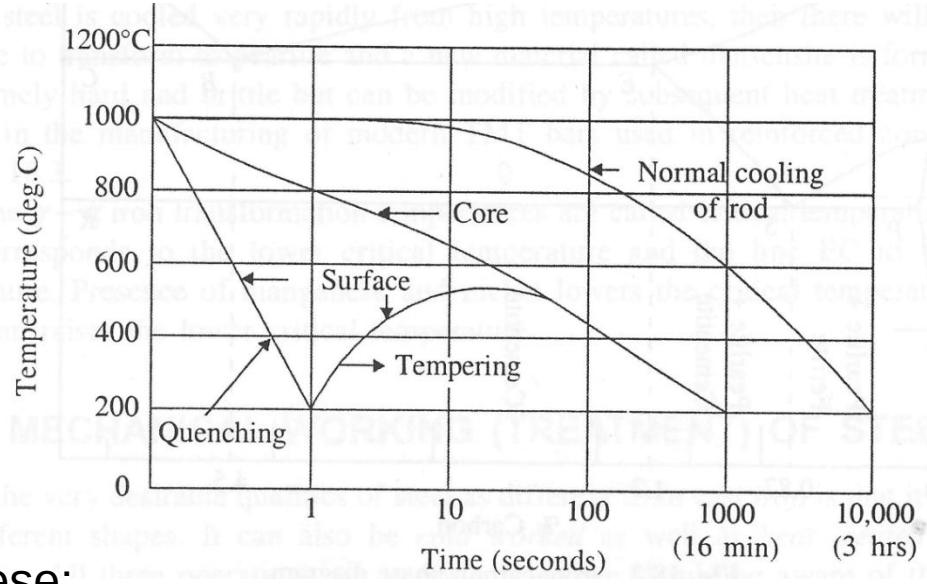
Thermomechanically Treated (TMT) or Quenched and Self-Tempered (QST) Bars

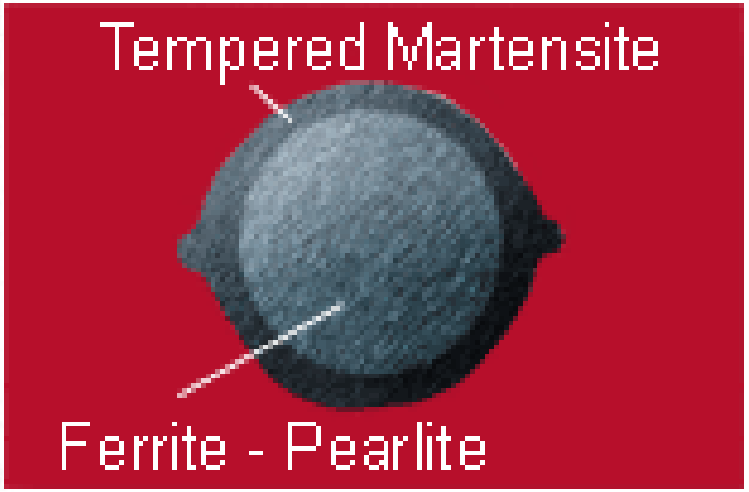
- After hot rolling to the desired size and shape, the low carbon steel bars are quenched with water and then cooled.
- Quenching converts the surface layer to (hard) martensite while the core remains as austenite.
- As the bar cools, heat flows from the core to the surface layer turning it to tempered martensite. The core transforms to ductile ferrite-pearlite.
- Yield Strength = 520-600 N/mm^2 and Total Elongation = 18 - 30%.



Thermomechanically Treated (TMT) or Quenched and Self-Tempered (QST) Bars



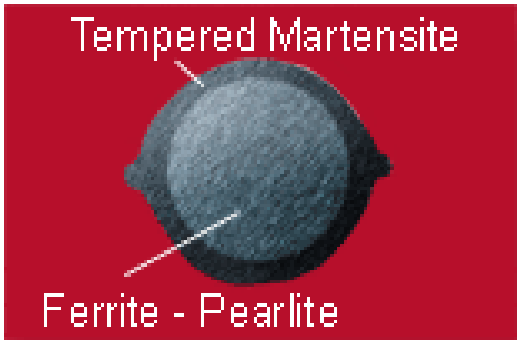




Varghese;

Thermomechanically Treated (TMT) or Quenched and Self-Tempered (QST) Bars

- **Identification/Selection of TMT/QST bars**
  - Select only of the hardened periphery and softer core are uniform and concentric in nature
  - Ideal if 20 to 30% of bar cross-sectional area is hardened periphery
  - Not ideal if >50% of bar cross-sectional area is hardened periphery (ductility reduces)
- **Advantages**
  - Low cost, high strength
  - High ductility (good for earthquake regions)
  - Elongation around 18-30%
  - Excellent weldability (due to lower carbon content)
  - No cold working (better corrosion resistance)
  - Uniformity



Corrosion Resistant Steel (CRS) TMT reinforcement

- Bars with small quantities of copper and chromium, and higher than usual percentage of phosphorus.
  - Carbon – 0.15%
  - Manganese – 1%
  - Sulphur – 0.04%
  - Phosphorous – 0.10%
  - Silicon – 0.45%
  - Corrosion resistant elements – 0.50% (minimum)
- Major producers - TATA & SAIL

Stainless steel reinforcement

- Stainless steels are alloys that contain at least 12% of chromium
- Other alloying elements such as nickel and molybdenum may also be present
- Chromium oxide layer forms on the stainless steel surface to prevent corrosion
- These steels offer good resistance to corrosion as long as the passive film can be maintained

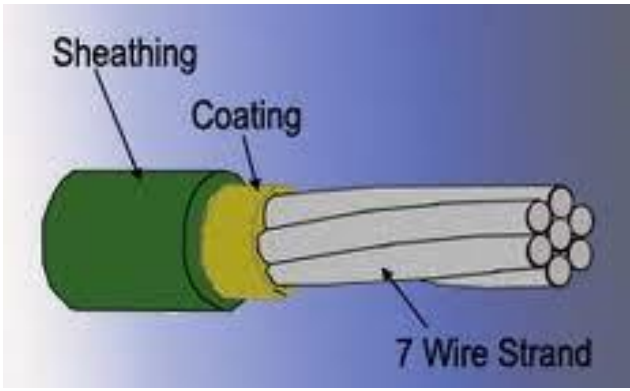
Prestressing steel reinforcement – 7-wire strands

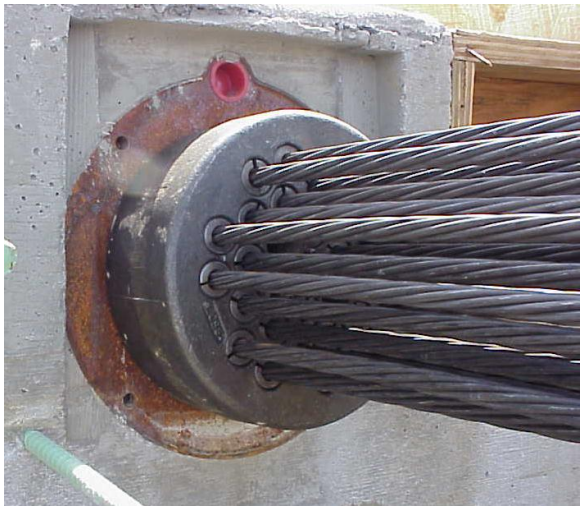
- High strength (1860 MPa)
  - Approximately 4.5 times more!
- Low relaxation
  - 1000 hour stress relaxation is typically less than 3.5%
- Necessary for long span construction
  - Cable stayed, suspension, prestressed concrete applications
- Unsheathed and sheathed strands





▲ Prestressing steel strand





Epoxy-coated steel reinforcement

- Can be a straight and bend bars
- Bars must be bent before epoxy-coating
- Check for “holidays” on the bar surface
- Provide multiple layers of epoxy coating
- Fusion bonded epoxy coating may be alternative



Significant corrosion can occur if the epoxy-coating is damaged/scratched





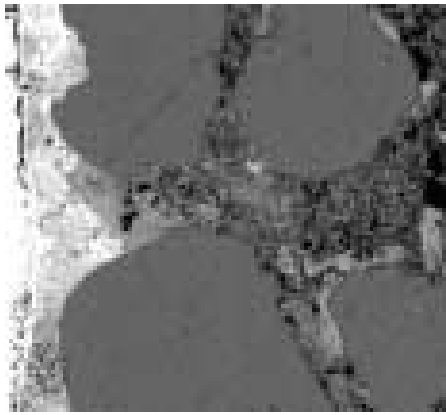
It is more dangerous to
use damaged epoxy-coated steel than
conventional uncoated steel

Courtesy: M. Thomas

Galvanized steel reinforcement



- Initial stable corrosion product
  - Calcium hydroxyzincate (at pH < 13.3)
- Zinc's corrosion products are loose, powdery minerals that are less voluminous than iron corrosion products and are able to migrate away from the galvanized rebar surface into the adjacent concrete matrix.



- As a result, corrosion of the zinc coating causes very little physical disruption to the surrounding concrete.

Galvanized steel reinforcement



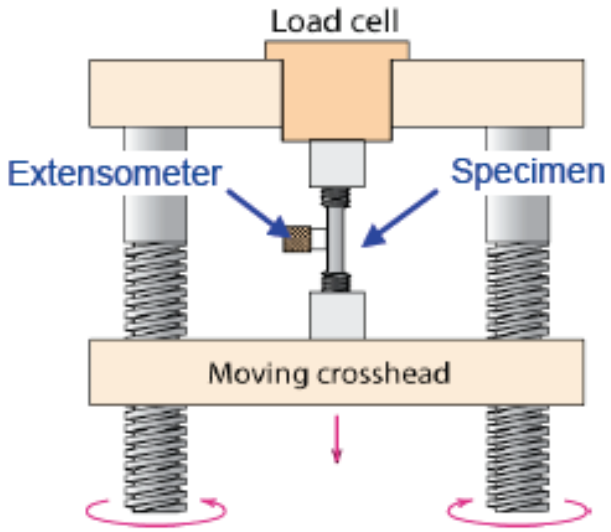
- Substantially higher chloride threshold (2-4 times) for zinc coating
- zinc has a much greater pH passivation range than steel, making galvanized rebar resistant to the pH lowering effects of carbonation as the concrete ages.
- Even when the zinc coating does start to corrode, its corrosion rate is considerably less than that of uncoated steel.



Tension test

- Many of the parameters measured depend not only upon material properties, but also upon the type of test and properties of the testing machine.

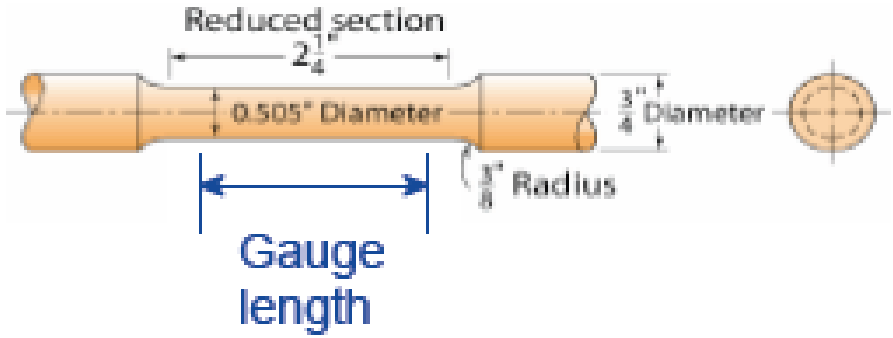
Typical Tensile Tester



STRESS-STRAIN DIAGRAM

Adapted from Fig. 6.3, *Callister 6e.*

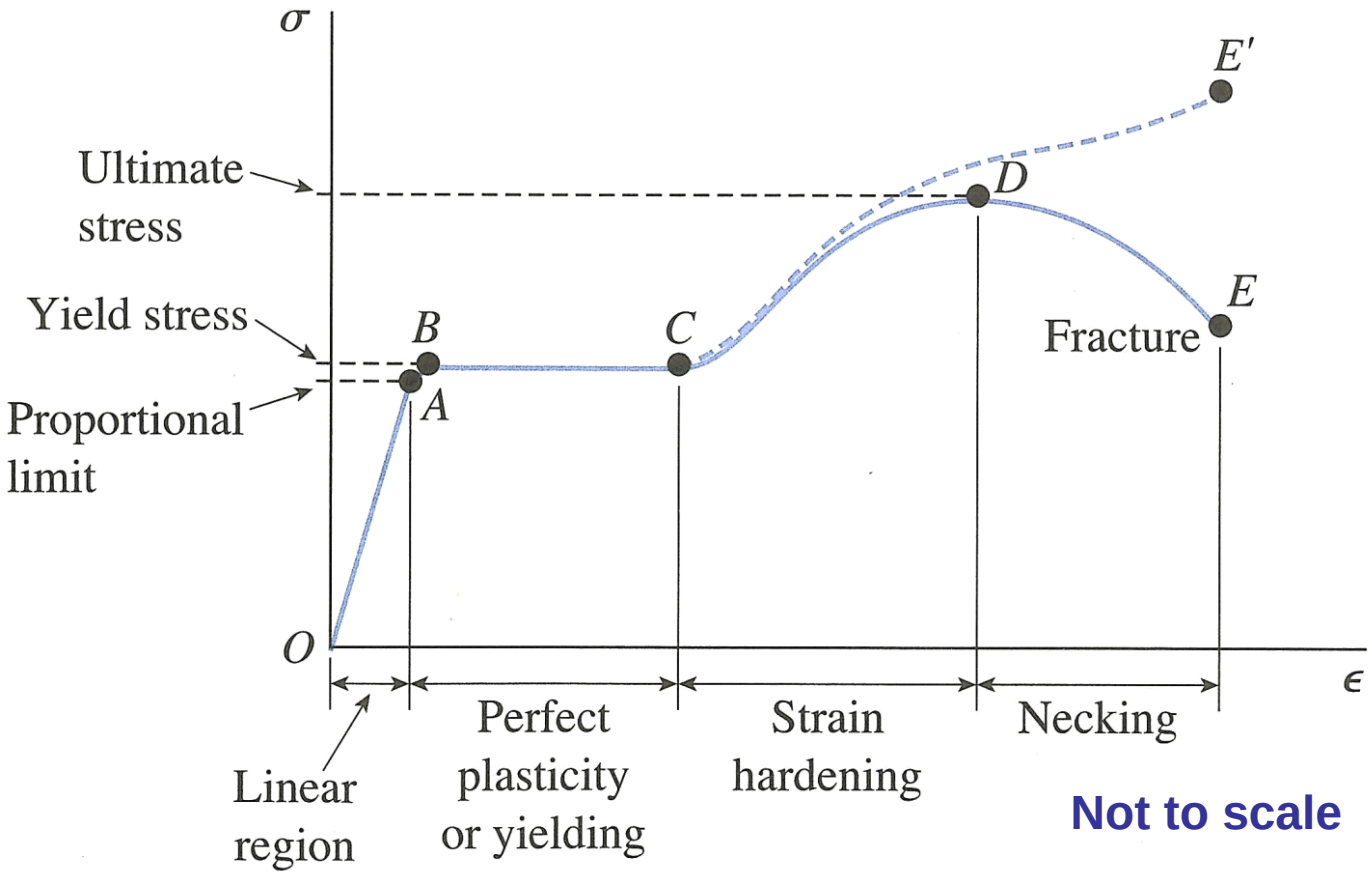
Typical Tensile Specimen

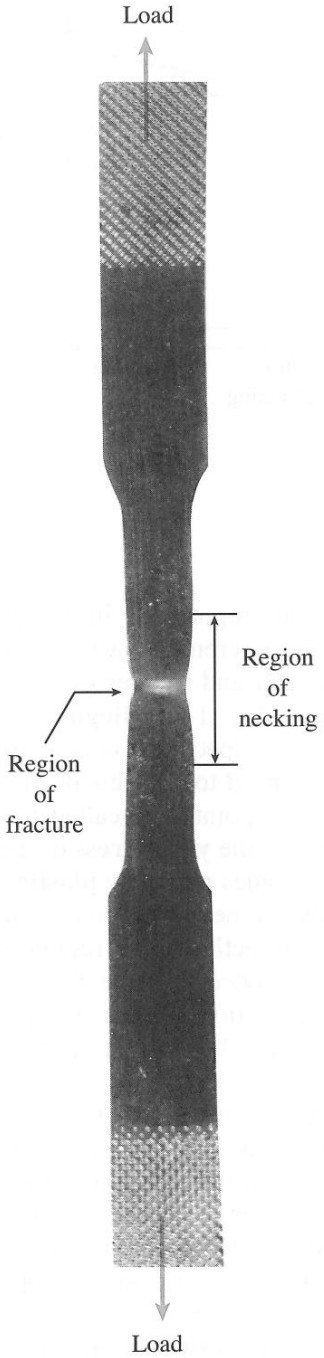


apted from Fig. 6.2, *Callister 7e.*

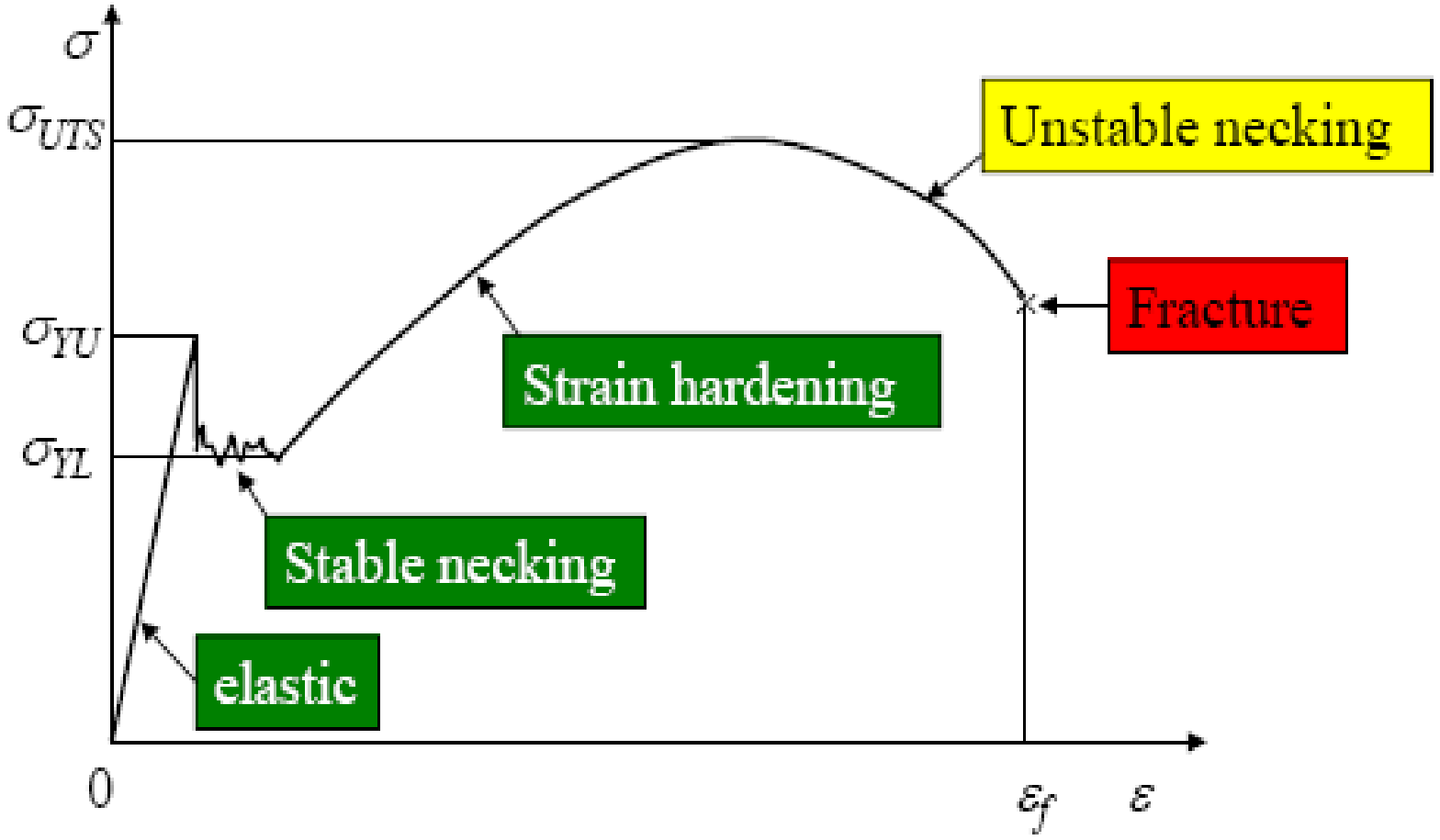
Stress-Strain Diagram (in Tension)

- When the initial area is used in the stress calculation, it is called nominal stress.
- When actual area of bar is used, it is the true stress.

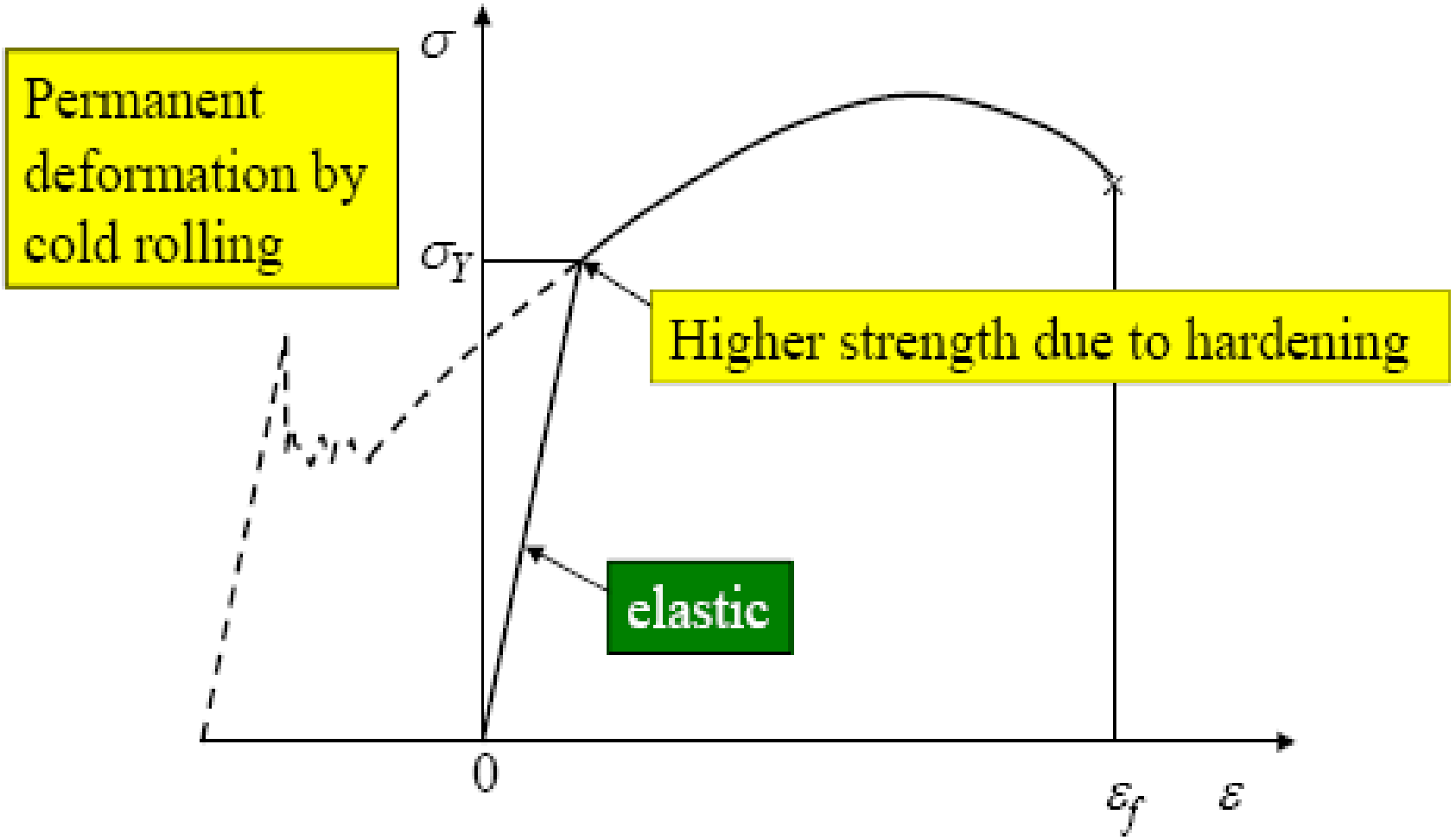




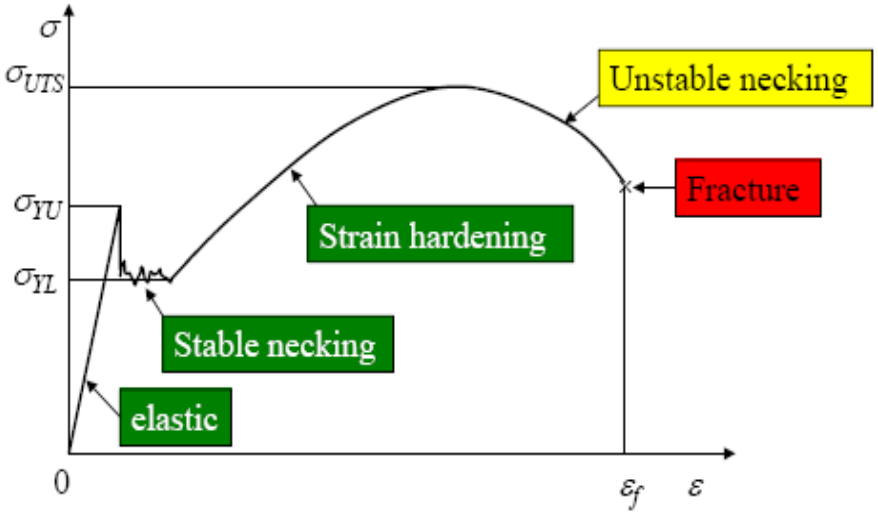
Difference between the behavior of hot-rolled steels



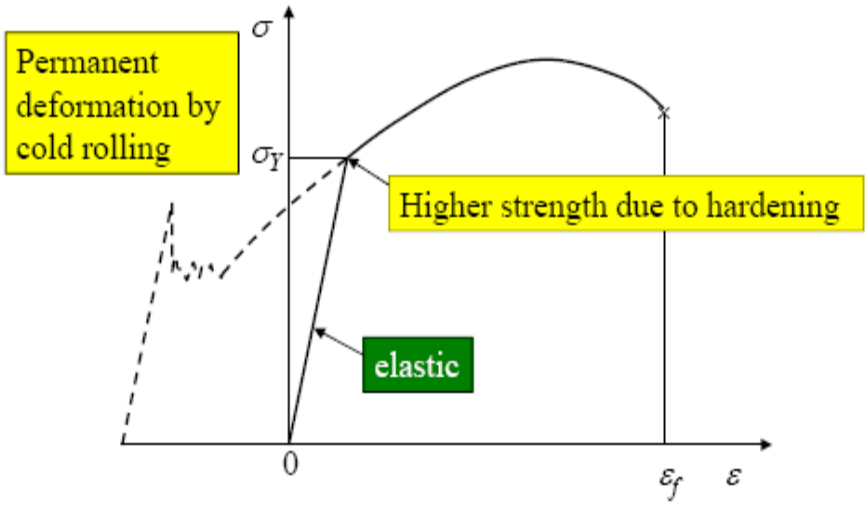
Difference between the behavior of cold-rolled steels



Difference between the behavior of hot-rolled and cold-rolled steels

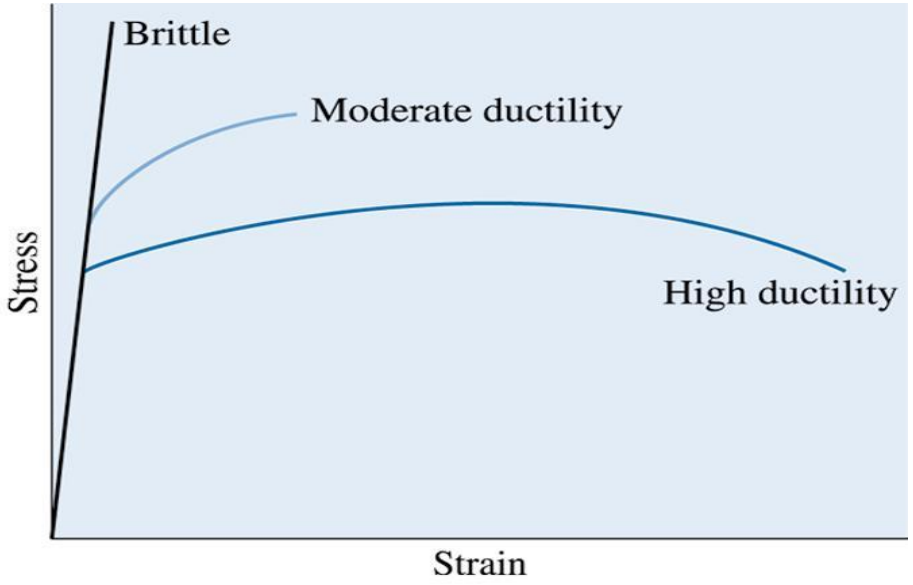


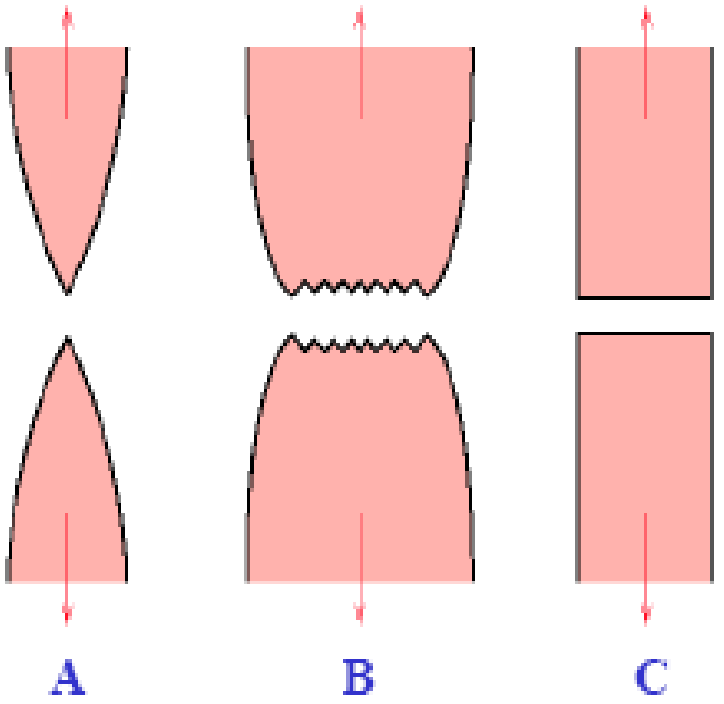
Hot rolled steel



Cold rolled steel

Ductility and Brittleness





A. **Very ductile**, soft metals (e.g. Pb, Au) at room temperature, other metals, polymers, glasses at high temperature.

B. **Moderately ductile fracture**, typical for ductile metals

C. **Brittle fracture,** cold metals, ceramics.

(c)2003 Brooks/Cole, a division of Thomson Learning, Inc.

Ductility and Brittleness

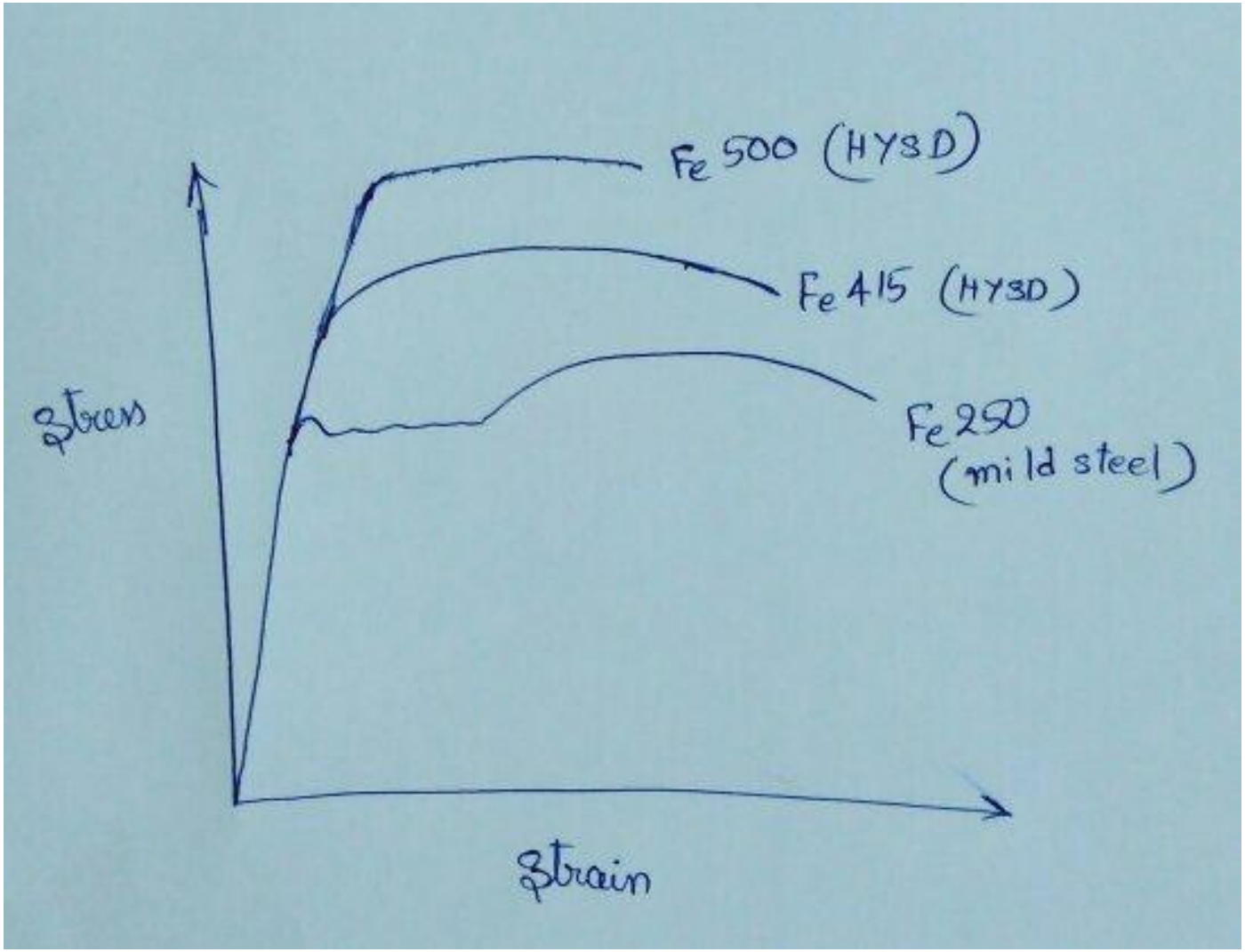
Ductile failure of Aluminum



Brittle failure of Mild steel



Ductility and Brittleness

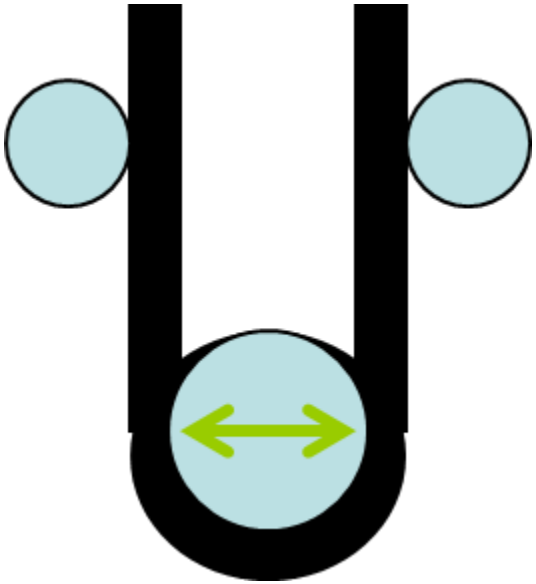


Bend Test (IS 1608) - Specimen considered to pass the test, if there is no crack or rupture

Table 4 Mandrel Diameter for Bend Test
(*Clause* 9.3)

Sl No.	Nominal Size mm	Mandrel Diameter for Different Grades						
		Fe 415	Fe415D	Fe500	Fe500D	Fe550	Fe550D	Fe600
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	Up to and including 20	3 ϕ	2 ϕ	4 ϕ	3 ϕ	5 ϕ	4 ϕ	5 ϕ
ii)	Over 20	4 ϕ	3 ϕ	5 ϕ	4 ϕ	6 ϕ	5 ϕ	6 ϕ

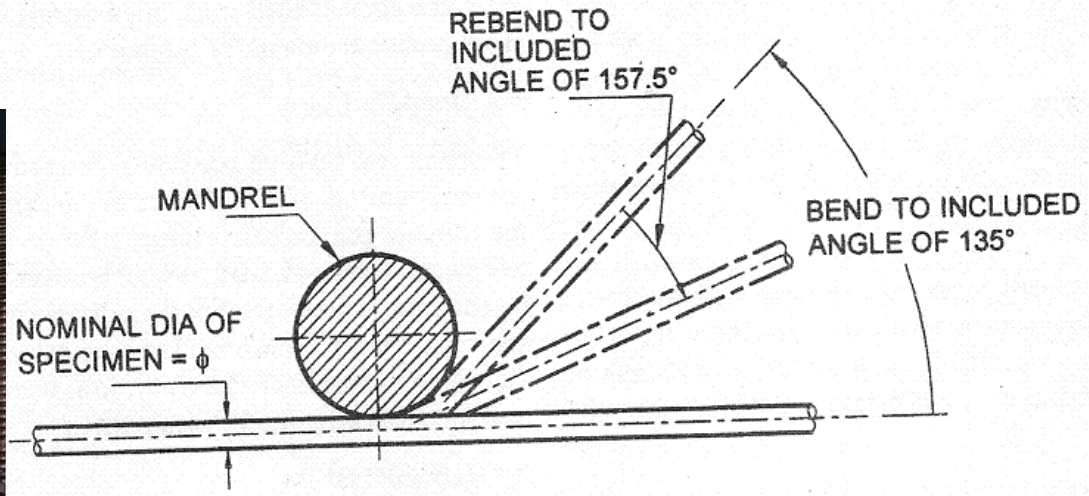
where ϕ is the nominal size of the test piece, in mm.





Re-bend test - Mandrel Diameter





NOTE—ϕ Represents the nominal size in mm of the test piece.

FIG. 2 REBEND TEST

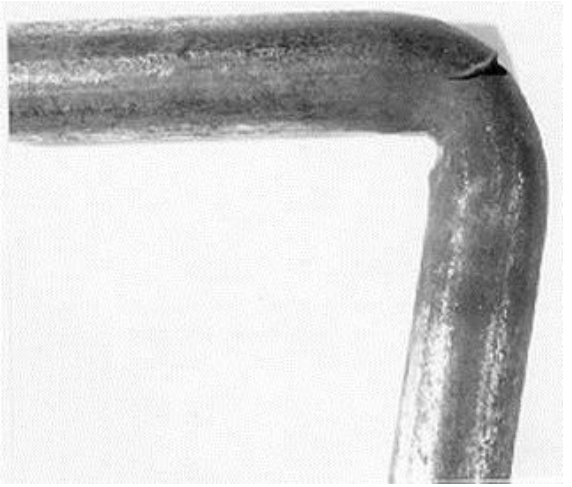
Sl No.	*Nominal Size of Specimen*	*Dia of Mandrel for Fe 415 and Fe 500*	*Dia of Mandrel for Fe 415D and Fe 500D*	*Dia of Mandrel for Fe550 and Fe 600*	*Dia of Mandrel for Fe 550D*
(1)	(2)	(3)	(4)	(5)	(6)
i)	Up to and including 10 mm	5 ϕ	4 ϕ	7 ϕ	6 ϕ
ii)	Over 10 mm	7 ϕ	6 ϕ	8 ϕ	7 ϕ

where ϕ is the nominal size of the test piece, in mm.

Re-Bend test

- The test piece bent to an inclined angle of 135°
- Mandrel diameter as given in the Table (Different from bend test) previous slide
- The bent piece should be aged for 30 min in boiling water
- Piece should be bent back to 157.5 °
- The purpose of re-bend test is to measure the effect of strain ageing on steel.
- Strain ageing has embrittlement effect which takes place after cold deformation by diffusion of nitrogen in steel

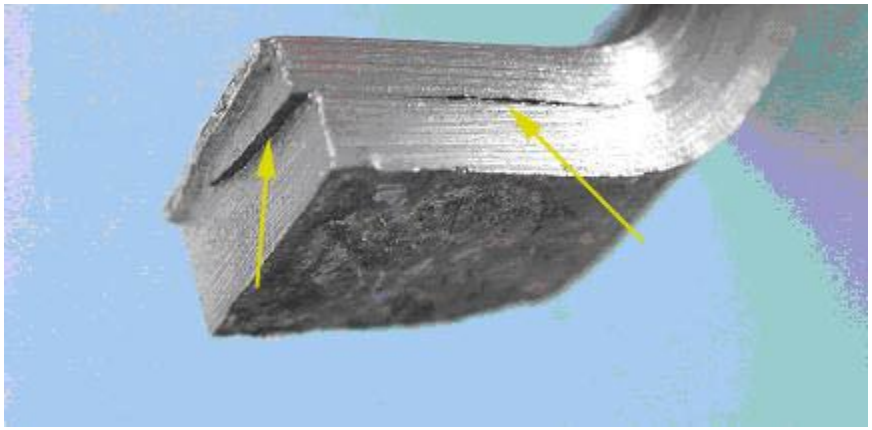




Re-Bend test

- Two further samples in respect of each failure
- If both additional test pass- the test is deemed to have passed
- Even if one out of two additional test fail, the specimen is considered to be a failure





Prefabricated steel structures

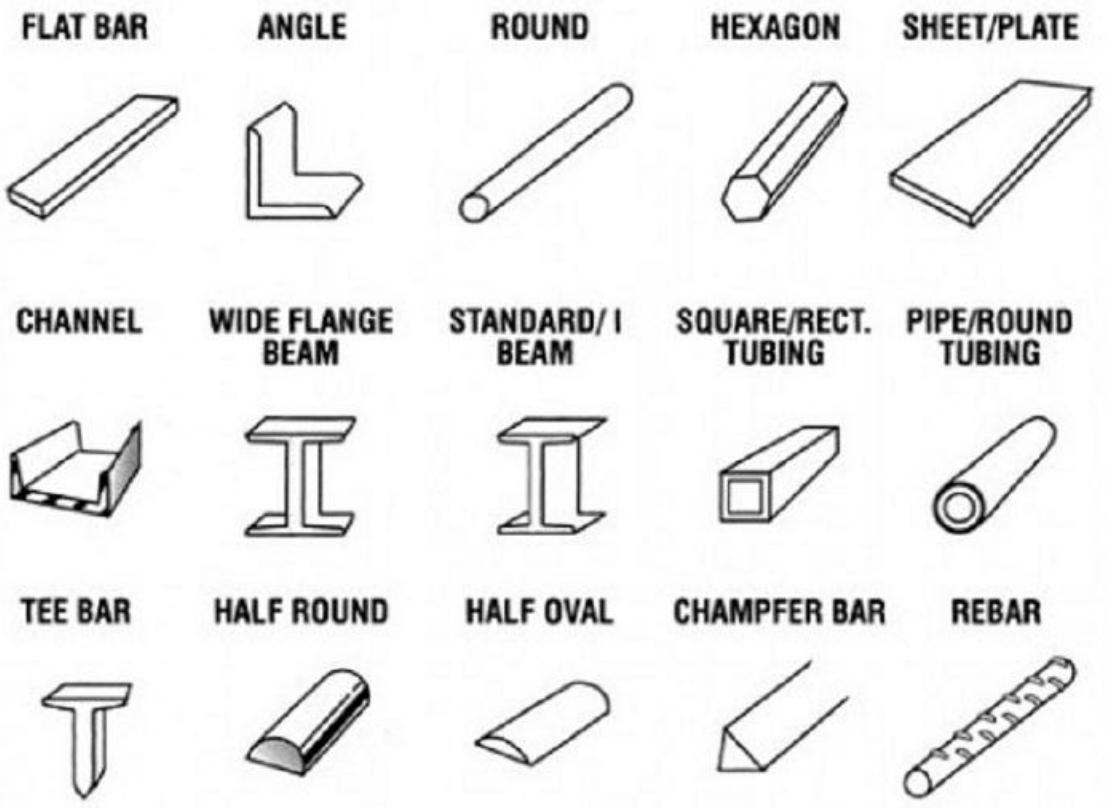


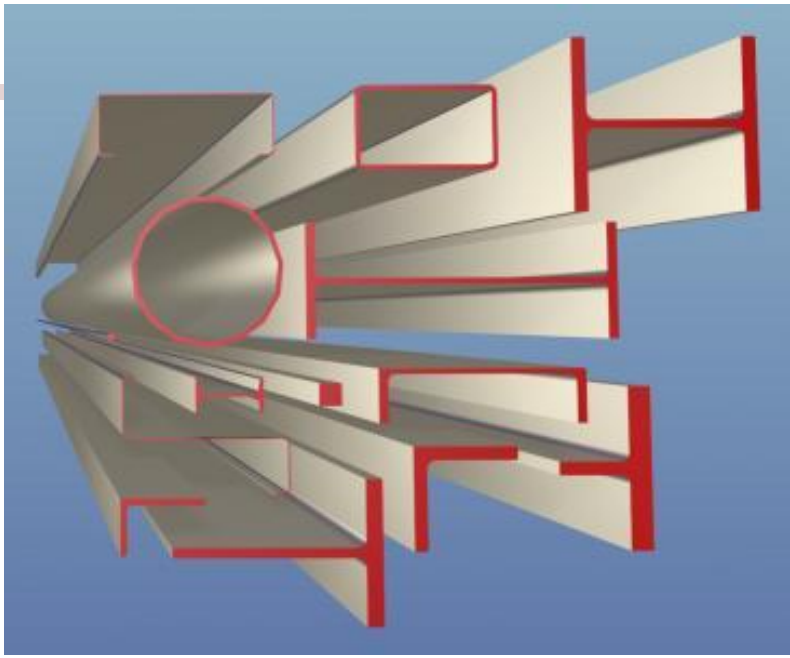






Various shapes and sections are used for building works







Plates:

- Plates may be of any size or thickness.
- Common uses of plates in building construction are as webs and flanges of deep beams, column flanges, column base, etc.

Flats:

- These are rolled as in the case of plate but are much longer in lengths and have shorter widths.
- The widths vary form 18mm to 50 mm. The minimum and maximum thickness vary from 3mm to 80 mm.
- Flat section may have one rounded side with greater thickness than the remaining section and this is called bulb ball. T

Angles:

- Angle sections are widely used in steel trusses. Most common types are angles with equal legs and with unequal legs and are designated by ISA width and height of legs of angles.
- The equal angles vary form 20 mm x 20mm to 200mm x 200mm in size and 3mm to 5mm in thickness.
- The term 20mm x 20mm denotes that the width of legs is 20mm overall. Unequal angles vary in size from 20mm x 30mm to 220mm x 100mm and thickness from 4mm to 20mm.
- Special angles with a square toe, round backed, acute, square root and bulb types are also available.



T-Sections:



- These are used for roof trusses and for certain built up columns.
- They are designated by the width of the stem, width of the table and by the thickness.
- The standard sizes vary form 40mm x 40mm to 150mm x 150mm with thickness from 6mm to 8mm.
- Special T-Sections with bulbs etc. are also used to some extent.

Channels:



- Channels are mainly used for beams, columns and top and bottom chord members of truss.
- They are designated by the depth, flange width and weight per unit length.

Channels contd…..

- The size varies from 50mm x 75mm x 3kg to 420mm x 100mm x 30kg.
- Whenever stronger channels of lesser depth are required, these are specially cast with greater thickness than given by the standards.

- Indian Standard Junior Channels (ISJC)
- Indian Standard Light Channels (ISLC)
- Indian Standard Medium Weight Channels (ISMC)
- Indian Standard Special Channels (ISSC)

Miscellaneous Sections:

- Z-Sections, rail-sections, troughs, bars etc. are used to a limited extent in steel work for a building.

Joists:

- Rolled Steel Joists or I-Sections are most commonly used for beams and columns.
- The British Standard Joists Section varies in size form 7 cm x 4cm to 60 cm x 17 cm. In U.S.A. I-sections have vary wide flanges are used.
- ISI handbook gives the following five series of Beam Sections.

- Indian Standard Junior Beams (ISJB)
- Indian Standard Light Beams (ISLB)
- Indian Standard Medium Weight Beams (ISMB)
- Indian Standard Wide Flange Beams (ISWB)
- Indian Standard H Beams (ISHB)



Video on Steel manufacturing